

System architecture

Business and technical design, end to end

Technical + business documentation | 19 September 2026

Business perspective

LangChat proves the YouTube LangChain diagram: chatbot + knowledge + chat history in a real DATABASE. Convex Free is that database. Python LangChain is the brain.

The problem (why this exists at all)

Lab memory died on process exit. Buyers of custom AI chat ask 'will it remember me?' Without durable storage the answer is no.

Who the customer is

DMB uses this as a demo product. End users are visitors chatting in a browser session.

How money is supposed to move

Not SaaS-billed yet. Demo value: show permanent memory on \$0 Convex + Render + Vercel free tiers.

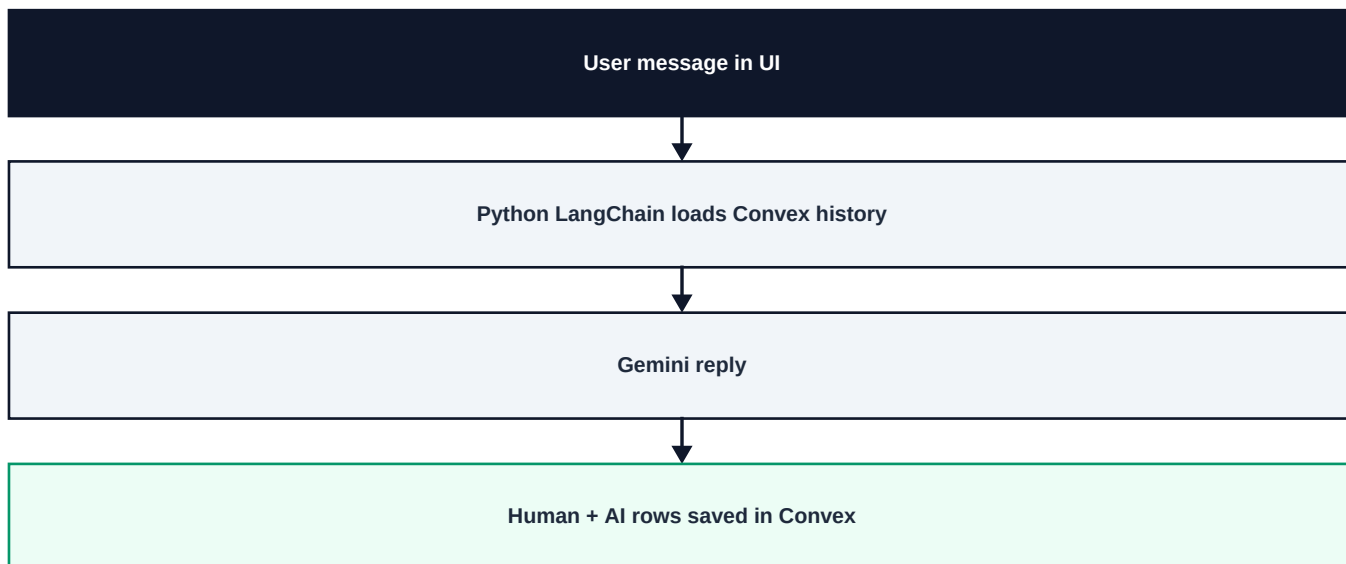


Figure B1. Durable memory loop.

How the business runs day to day

Step	In the real world	What this software is for
Capture	User types a message.	Web POST /chat
Recall	API loads last N messages.	messages:listByThread
Think	LangChain invokes Gemini.	ChatGoogleGenerativeAI

Remember	Append human + AI.	messages:append + thread title
Show	UI updates live.	Convex useQuery subscription

What 'winning' looks like (KPIs)

- Refresh keeps history.
- Free-tier Convex stays under 1M calls / 0.5 GB for demo traffic.
- Cold Render wake still returns a reply (visitor may wait once).

1. What this product is

DMB LangChat is a standalone sibling of dmbportfolio, CRM, and LMS. It is not tutoring and not a sales CRM. It is a LangChain chat with Convex-backed threads.

2. Live system map

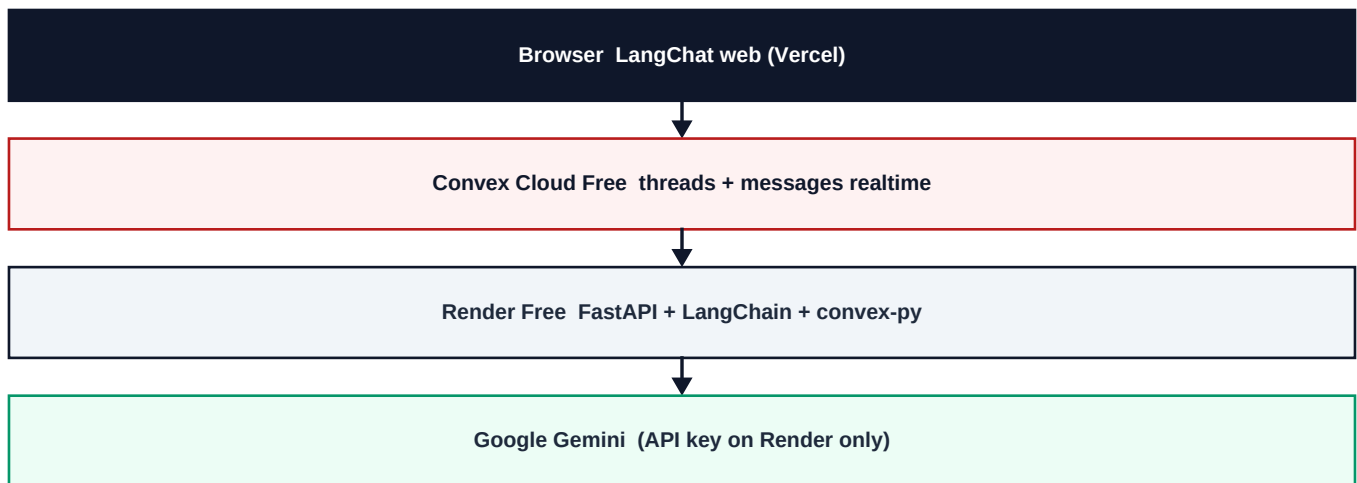


Figure 1. Three free hosts. Marketing site is not in the path.

3. Data model

Table	Fields	Indexes
threads	sessionId, title, updatedAt	by_session
messages	threadId, role, content, createdAt	by_thread

Roles: system | human | ai. First human message can rename title from 'New chat'.

4. API surface

Method	Path	Purpose
GET	/health	Convex + model config flags
POST	/threads	Create thread for sessionId
POST	/chat	Append human, invoke model, append AI

5. Key files

Area	Path
Schema	convex/schema.ts
Threads / messages	convex/threads.ts, convex/messages.ts
API	api/main.py
UI	web/src/App.tsx
Docs kit	Documentations/_pdf_kit.py

6. Free-tier limits to watch

- Convex Free: hard caps (function calls, storage).
- Render Free: sleeps; first chat after idle is slower.
- Vercel Hobby: fine for a static/Vite SPA.
- Never put GOOGLE_API_KEY in the web bundle.

7. Out of scope

- RAG / vector search, agent tools, payments, marketing SSO hub merge.

End of document. See also SSO/Auth and User Guide PDFs in this folder.